



SARUS AEROSPACE
PRIVATE LIMITED · INDIA

USER MANUAL

Sarus Operation Planner

Ground control station for ArduPilot aircraft. Installation, specifications, and a complete guide to every screen, feature and keyboard shortcut.



SOFTWARE VERSION
1.1.0

PLATFORM
Windows 10 / 11

DOCUMENT
SOP-UM-001

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01 Safety and about this manual

SAFETY FIRST

Take every propeller off before you calibrate, test motors, change parameters on the bench or install firmware. One wrong parameter is enough to crash an aircraft, so save a backup before each session and change one thing at a time. Even during an AUTO mission, keep the radio control transmitter in your hands, ready to take over. Fly only where and how the law allows; in India that means the Drone Rules 2021 and the DGCA's Digital Sky requirements. The software comes without warranty under the GNU GPL v3, and the operator is responsible for every flight.

This manual covers Sarus Operation Planner 1.1.0 for Windows. The program is built on ArduPilot Mission Planner 1.3.83 and keeps every one of its screens, controls and keyboard shortcuts, so Mission Planner tutorials apply here too. What differs is the look, the way parameters are edited and protected (chapter 11), the airframe limits and in-flight alarm (chapter 11), more reliable communication with the aircraft, and where files are kept.

Conventions: on-screen names are in **bold**. Menu paths are written **CONFIG → Full Parameter List**. Keys are shown as **Ctrl** + **P**. Parameter names are in MONOSPACE.

02 Product overview

Sarus Operation Planner is a ground control station (GCS): Windows software that connects to a drone's flight controller and lets you set up, plan, fly, monitor and analyse ArduPilot aircraft.

What you can do with it

Before a flight you can install or update the firmware on the flight controller, calibrate the accelerometers, compass, radio and ESCs, set flight modes and failsafes, and tune the aircraft: PIDs, TECS, navigation and the VTOL settings, down to any single parameter. Missions are planned on a map with waypoints, surveys, geofences and rally points.

In the air, the DATA screen gives you a live HUD, map, gauges and quick actions, and the program can speak mode changes, altitude and battery warnings. Afterwards you can replay the telemetry, download the flight controller's own log and analyse both. ArduPilot's simulator lets you fly without hardware, and optional equipment such as GPS, airspeed sensors, rangefinders, gimbals, OSD, CAN devices, ADS-B receivers and telemetry radios is set up from the same program.

Supported vehicles

VEHICLE	FIRMWARE	SARUS TEST STATUS IN 1.1.0
Fixed wing and QuadPlane (VTOL)	ArduPilot Plane	Tested in the simulator
Multirotor and helicopter	ArduPilot Copter	Multirotor tested in the simulator
Rover and boat	ArduPilot Rover	Tested in the simulator
Submarine, antenna tracker	ArduSub, AntennaTracker	Supported as in Mission Planner, not tested by Sarus

Flight controllers: any board that runs ArduPilot, including Pixhawk-class boards (Pixhawk, Cube, Holybro, Matek, CUAV and others).

What is different from Mission Planner

Look	Sarus Dark (default), Sarus Light and the opt-in Sarus Glass themes, Sarus menu icons and the "Powered by Sarus Aerospace" badge. HUD and map colours are unchanged.
Parameters	Values are taken exactly as typed: no recommended ranges, no clamping, no suggested parameter sets. Read-only values can be changed after confirmation. Every written value is re-read from the aircraft (chapter 11).
Parameter lock	Setup changes need the owner's admin password, enforced by the program and by aircraft running Sarus firmware (11.3).
Airframe limits and alarm	Values beyond what the airframe can do blink red, and a siren asks continue, hold or return when a mission is not going as set (11.4, 11.5).
Reliability	Replies from the aircraft can no longer be lost when two requests wait at the same time (parameters, commands, missions).

Coexistence	Installs beside Mission Planner with separate settings, logs, maps and updates.
Privacy	Anonymous usage statistics are off by default.
Airspace	No Altitude Angel or other airspace service is included, so the program never restricts where you fly. The aircraft's own geofence and failsafes still apply, under your control.
Network safety	The built-in web server accepts flight commands only from this PC; other devices on the network can still view telemetry.

03 Specifications and requirements

Software

Product	Sarus Operation Planner
Version	1.1.0 (file version 1.3.83, the Mission Planner base)
Sarus firmware	Sarus-2 on ArduPilot 4.7.1 and on ArduPilot 4.6.3 (Plane with QuadPlane, Copter, Rover)
Publisher	Sarus Aerospace Pvt Ltd, India · sarusaerospace.in
Based on	ArduPilot Mission Planner 1.3.83
Communication protocol	MAVLink 1 and 2
Connections	USB/serial (COM ports), TCP, UDP, UDP client, WebSocket
Installer	Windows Installer package (.msi), about 110 MB
Licence	GNU General Public License v3, source at github.com/LikhitS/SarusOperationPlanner
Languages	English, plus the Mission Planner translations (CONFIG → Planner → UI Language)

Computer requirements

ITEM	MINIMUM	RECOMMENDED
Operating system	Windows 10 (64-bit)	Windows 11 (64-bit)
.NET Framework	4.7.2 (built into Windows 10 version 1803 and later)	4.8
Processor	Dual-core, 2 GHz	Quad-core or better
Memory	4 GB	8 GB or more
Disk	500 MB plus map cache and logs	5 GB or more free for maps and logs
Display	1280 × 720	1920 × 1080, 100% scaling
Ports	One USB port for the flight controller or telemetry radio	

ITEM	MINIMUM	RECOMMENDED
Internet	Optional. Needed for firmware downloads, parameter descriptions, online maps and updates. Flying works offline with cached maps.	

Above 100% Windows display scaling, text looks slightly soft. This is the same in Mission Planner.

04 Installation

4.1 Download

Download the latest SarusOperationPlanner-x.y.z.msi from the Sarus releases page: github.com/Likhits/SarusOperationPlanner/releases, or from the Sarus Aerospace website.

4.2 Install

- 1 Double-click the downloaded .msi file.
- 2 If Windows shows **Windows protected your PC**, click **More info**, check that the file is the one from Sarus, then click **Run anyway**. This appears until the installer carries a code signature.
- 3 Read and accept the licence (GNU GPL v3), then click **Next**.
- 4 Keep the default folder C:\Program Files (x86)\Sarus Operation Planner or choose another, then click **Install** and approve the administrator prompt.
- 5 When Windows asks to install device drivers (for flight-controller USB ports), click **Install**. These are the same drivers Mission Planner installs.
- 6 Click **Finish**. Start the program from the Start menu: **Sarus Operation Planner**.

SILENT INSTALL (IT DEPARTMENTS)

msiexec /i SarusOperationPlanner-1.3.83.msi /qn /norestart installs without questions. In a silent install, Windows adds only the USB drivers whose publisher is already trusted; ArduPilot boards on Windows 10/11 normally use the built-in Windows USB serial driver. Uninstall with `msiexec /x` and the same file, or from Windows Settings → Apps.

4.3 Next to Mission Planner

Sarus Operation Planner and Mission Planner can be installed on the same PC. Each has its own program folder, Start-menu entry, settings, logs, map cache and update channel. Installing or removing one does not affect the other. Only one program can use a given COM port at a time.

4.4 Updating

On start, the program checks the Sarus releases on GitHub. When a new version exists, it offers to update. The updater downloads only the files that changed and verifies each file's checksum. You can also check manually in **HELP → Check for Updates**, or install a newer .msi over the current one; settings are kept.

4.5 Uninstalling

Windows Settings → Apps → Installed apps → **Sarus Operation Planner** → Uninstall. Your logs, missions and settings in Documents\Sarus Operation Planner are kept; delete that folder by hand if you no longer need them.

05 First start and the main window

5.1 Questions on first start

MESSAGE	WHAT TO DO
Windows Security: allow network access	Not shown after a normal installation: the installer already allows the program on every network type, and removes that permission when you uninstall. If you run the portable copy, Windows asks once; tick both boxes and click Allow .
Restart to enable a plugin (for example Drone ID)	Restart the program if you want the plugin, or disable it in Ctrl + P if you do not.
Other "show again?" messages	Informational. Tick "don't show again" if you prefer.

5.2 The main window

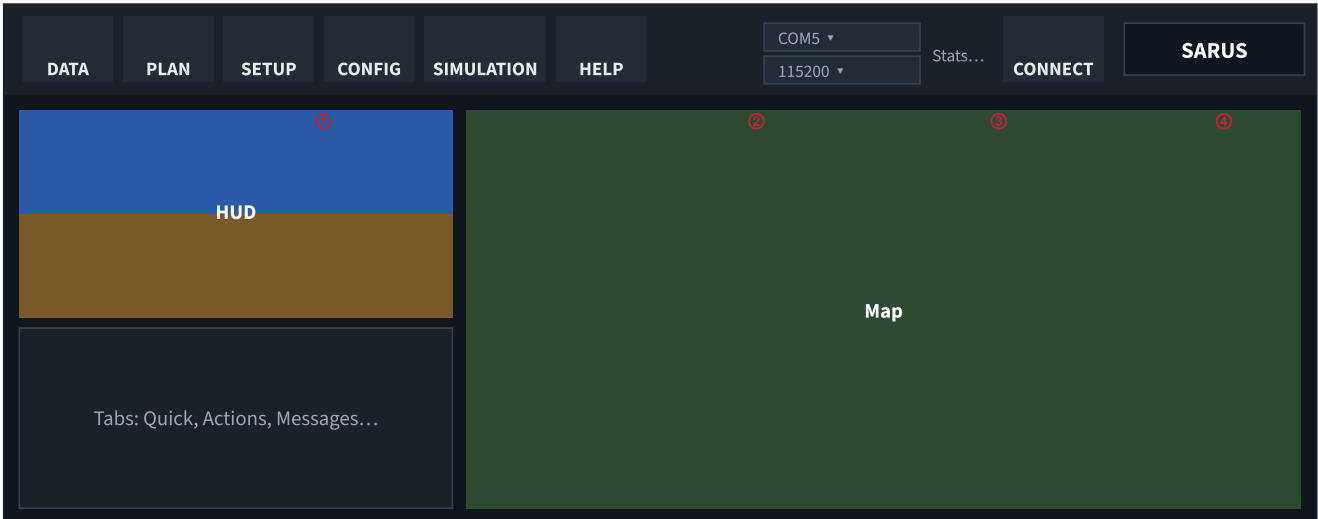


Figure 1. Main window on the DATA screen. ① Screen menu ② Connection type and speed ③ Link statistics and CONNECT ④ Sarus badge (opens sarusaerospace.in). Version 1.1.0 shows "Powered by Sarus Aerospace" in the badge.

MENU	SCREEN	USE IT TO	KEY
DATA	Flight Data	Monitor and command the aircraft in flight; replay logs	F2
PLAN	Flight Planner	Create missions, geofences and rally points	F3
SETUP	Hardware Config	Install firmware, calibrate, set up hardware	
CONFIG	Software Config	Tune, edit parameters, program settings	F4

MENU	SCREEN	USE IT TO	KEY
SIMULATION	Simulation	Fly a simulated aircraft (SITL)	
HELP	Help	Updates, change log, console window	

The program remembers its window size, theme and last connection.

06 Connecting to an aircraft

6.1 Connection types

TYPE (TOP-RIGHT LIST)	USE FOR	USUAL SPEED / PORT
COMx	USB cable to the flight controller, or a USB telemetry radio (SiK, RFD900)	USB: 115200 · SiK radio: 57600
AUTO	Tries the available COM ports and speeds automatically	–
TCP	Network link to an aircraft, a companion computer or SITL	e.g. 127.0.0.1:5760
UDP	Listen for an aircraft broadcasting telemetry (Wi-Fi radios, Herelink, SITL)	14550
UDPCl	Send to a fixed address and port (UDP client)	host and port
WS	WebSocket link	URL

6.2 Connect

- 1 Plug in the flight controller (USB) or the ground telemetry radio, and power the aircraft.
- 2 Choose the COM port (or TCP/UDP) and the speed in the top-right lists.
- 3 Click **CONNECT** or press **F12**.
- 4 The program waits for a heartbeat, then downloads all parameters (a progress window shows this). When it finishes, the HUD moves and the title bar shows the firmware version.

Click **DISCONNECT** (or **F12** again) to disconnect. **Ctrl + T** connects without downloading the parameters (advanced). **Stats...** under the connection lists shows link quality, bytes per second and packet loss.

With several aircraft or components on one link, choose which one to control from the system-ID list that appears in the connection area.

07 DATA – Flight Data screen

This is the screen you fly from. The HUD sits at the top left, the tabs below it and the map on the right; drag the dividers to change the proportions.

7.1 HUD (head-up display)

The HUD draws the artificial horizon with roll and pitch, plus heading, airspeed and ground speed (left tape), altitude (right tape), vertical speed, flight mode, armed state, GPS status, battery, link quality, distance to waypoint and current waypoint, plus warnings such as PreArm messages, EKF problems and low battery. HUD colours are those of Mission Planner.

Right-click the HUD for: **Video** (start a camera, MJPEG or GStreamer source, Herelink video, record HUD to AVI), **Set Aspect Ratio**, **User Items** (add any telemetry value to the HUD), **Swap With Map**, **Ground Color**, **Battery Cell Voltage**, **Show icons** and **Russian Hud**.

7.2 Tabs

Press **Ctrl** + **1** to **Ctrl** + **0** to jump to the 1st to 10th visible tab. Right-click a tab strip to choose which tabs are shown.

TAB	CONTENTS
Quick	Large-number display of selected values (altitude, speed, distance, yaw, vertical speed...). Double-click a value to choose another. Right-click to set the number of values, or undock.
Actions	Do Action (any command) · Set WP · Set Mode · Set Mount · Arm/Disarm · Auto · Loiter · RTL · Restart Mission · Resume Mission · Abort Landing · Set Home Alt · Clear Track · Raw Sensor View · Joystick · Message · Change Speed · Change Alt
Messages	All text messages from the aircraft, with time stamps (PreArm, EKF, mode changes, errors)
Actions (simple)	Reduced set of large action buttons
PreFlight	Configurable pre-flight checklist with live checks
Gauges	Analogue speed, altitude, vertical-speed and heading gauges
Transponder	ADS-B transponder control: squawk, flight ID, IDENT, mode (STBY / ON / ALT)
Status	Every telemetry value the program receives, in one list
Servo/Relay	Toggle relays and drive servo outputs (low / high / toggle)
Aux Function	Trigger ArduPilot auxiliary functions (as an RC switch would)
Scripts	Run Python scripts on the ground station
Payload Control	Gimbal roll, pitch and pan, video control, reset position
Telemetry Logs	Load and replay a .tlog (Play/Pause, speed 0.1× to 10×, slider), convert tlog to KML or graph

TAB	CONTENTS
DataFlash Logs	Download DataFlash Log Via Mavlink · Review a Log · Auto Analysis · Create KML + gpx · Convert .Bin to .Log · Create Matlab File · Geo Reference Images

7.3 Map

On the map you see the aircraft and its track, home, the mission, geofence, rally points, ADS-B traffic and the planned path. Scroll to zoom; drag to pan. Hold **Ctrl** and click the map to send the aircraft there (same as **Fly To Here**, guided mode).

Right-click the map: Fly To Here · Fly To Here Alt · Fly To Coords · TakeOff · Add Poi · Point Camera Here / Point Camera Coords · Trigger Camera NOW · Set Home Here · Set EKF Origin Here · Flight Planner · Camera Overlap · Jump To Tag. Checkboxes under the map control **Auto Pan**, **Tuning** (live graph of selected values) and the zoom level.

GUIDED COMMANDS

Fly To Here changes the aircraft to GUIDED mode. Check the altitude before you send it.

7.4 Log playback keys

While a telemetry log is loaded: **Space** plays or pauses, numeric-keypad **+** speeds up and **-** slows down.

08 PLAN – Flight Planner screen

Missions, geofences and rally points are created, loaded and sent from here. The map takes most of the screen, with the mission table along the bottom and the action panel on the right.

8.1 Making a mission

- 1 Check **Home Location** (filled in from the aircraft when connected).
- 2 Set the defaults: **WP Radius**, **Loiter Radius**, **Default Alt**, and the altitude frame (**Relative**, **Absolute** or **Terrain**). Tick **Verify Height** to use terrain data.
- 3 Left-click the map to add waypoints. Drag a waypoint to move it.
- 4 In the table, change each row's **Command** (WAYPOINT, VTOL_TAKEOFF, VTOL_LAND, LOITER_TURNS, DO_JUMP, DO_SET_SERVO, DO_CHANGE_SPEED...) and its parameters, latitude, longitude and altitude. The column headers change to explain the parameters of the selected command. Use the arrows to reorder and **X** to delete.
- 5 Click **Write** (**Ctrl** + **Shift** + **W**) to send the mission to the aircraft, then **Read** (**Ctrl** + **Shift** + **R**) to confirm it. **Write Fast** (**Ctrl** + **Shift** + **F**) sends without waiting for each item.
- 6 Save the mission with **Ctrl** + **S**; open one with **Ctrl** + **O**. **Ctrl** + **Z** undoes the last change.

The status bar shows total distance, distance between points, and an **Alt Warn** indication. Use **Elevation Graph** (right-click) to check terrain clearance.

8.2 Right-click menu on the map

Insert / Delete	Insert Wp · Insert Spline WP · Delete WP · Add Below
Loiter, Jump, RTL, Land, Takeoff	Add a loiter (Forever, Time, Circles), a jump (Start, WP #), RTL, Land or Takeoff command at the clicked point
DO_SET_ROI	Point the camera or vehicle at a location
Clear Mission	Remove all waypoints (on screen only until you Write)
Draw a Polygon	Draw, clear, save or load a polygon; from SHP or from current waypoints; offset; show its area
Geo-Fence	Upload, download, set return location, load from or save to file, clear; inclusion and exclusion fences
Rally Points	Set rally point, upload, download, save to or load from file, clear
Auto WP	Survey (Grid) · Create Circle Survey · Create Wp Circle · Create Spline Circle · Text
Map Tool	Measure Distance · Rotate Map · Zoom To · Prefetch (download map tiles for offline use) · Prefetch WP Path · KML Overlay · Elevation Graph · Reverse WPs · Modify Alt · Enter UTM Coord · GDAL opacity
File Load/Save	Load WP File · Load and Append · Save WP File · Load KML File · Load SHP File
POI	Add, delete, edit points of interest
Other	Set Home Here · Tracker Home · Switch Docking · Zoom to Vehicle / Mission / Home

Selecting several waypoints: hold **Ctrl** and drag a box, or **Ctrl** +click markers; then drag to move them together.

Geofence and rally editing: use the drop-down on the right of the screen (**MISSION** / **FENCE** / **RALLY**) to edit those lists with the same table and map tools. Fence points can be inclusion or exclusion zones (polygons or circles).

Map type: choose the map provider from the list on the right. **Ctrl** + **X** opens the map cache manager.

09 SETUP – Hardware setup

The pages are listed on the left. Those that need a connected aircraft stay greyed out until you connect.

9.1 Install Firmware

- 1 **Disconnect** (do not press CONNECT). Plug the flight controller into USB.
- 2 Open **SETUP** → **Install Firmware**. The page shows the latest release for each vehicle (Plane, Copter, Heli, Rover, Sub, Antenna Tracker). It reads ArduPilot's live firmware list together with the Sarus firmware list, so new releases of either appear the next time the program starts.
- 3 Click the vehicle (for a QuadPlane: **Plane**). For Copter, choose the frame type when asked. If more than one build exists for your board, a selection window lists them by file address: ArduPilot's, Sarus on 4.7.1 and Sarus on 4.6.3. The Sarus files are the ones whose names start with Sarus-. Installing firmware needs the admin password (11.3).
- 4 Follow the prompts (unplug and replug the board if asked). Wait for "Upload Done". Never unplug during flashing.

All Options lists every version and board. **Beta firmwares** offers test releases. **Load custom firmware** flashes a .apj file you built or were given. **Bootloader Update** / **Force Bootloader** update the board's bootloader (advanced). **Install Firmware Legacy** is the older version of this page.

9.2 Mandatory Hardware

Frame Type	Copter frame class and type (Quad X, Hexa...). For QuadPlane, set Q_FRAME_CLASS / Q_FRAME_TYPE in the parameter list.
Accel Calibration	Calibrate Accel (six positions: level, left, right, nose down, nose up, back), Calibrate Level, Simple Accel Cal
Compass	Priority order, enable/disable compasses, orientation, Onboard Mag Calibration (rotate the aircraft on all axes until done), Large Vehicle MagCal
Radio Calibration	Move all sticks and switches to their ends; Click when Done. Check each channel moves the right way.
Servo Output	Function, min, trim, max and reverse for each output
Serial Ports	Protocol and baud rate for each serial port
ESC Calibration	ESC calibration and motor PWM type (Copter)
Flight Modes	Six modes for the mode switch; current mode and PWM shown live; Save Modes
FailSafe	Radio, battery (low and critical voltage / mAh) and GCS failsafes and their actions; live RC input and output display
HW ID	Hardware IDs of the connected sensors
ADSB	ADS-B receiver setup

9.3 Optional Hardware

RTK/GPS Inject (RTCM corrections) · SiK Radio (configure SiK telemetry radios) · CubeID Update · CAN GPS Order · Battery Monitor and Battery Monitor 2 · DroneCAN/UAVCAN · Joystick · Compass/Motor Calib · Range Finder · Airspeed · PX4Flow · Optical Flow · OSD · Camera Gimbal · Antenna tracker · Motor Test · Bluetooth Setup · Parachute · ESP8266 Setup · FFT Setup.

MOTOR TEST

Props off. Set **Throttle %** (start at 5%) and **Duration**, then test each motor. Check the order and direction against your frame diagram. Set **Motor Spin Arm** and **Set Motor Spin Min** set the spin-when-armed and minimum-flying outputs.

9.4 Advanced

Terminal (text console to the board or a companion computer) and **Script REPL** (interactive Lua on the flight controller).

10 CONFIG – Software configuration and tuning

PAGE	CONTENTS
Flight Modes	Same as SETUP → Flight Modes
GeoFence	Fence enable, type, action, maximum altitude, radius
Basic Tuning	Simple sliders for responsiveness (Copter); key rates and limits (Plane)
Extended Tuning	Copter: rate and angle PIDs, navigation. Plane: servo roll/pitch/yaw PIDs, L1 turn control, TECS (climb max, sink min, sink max, time constant, damping), airspeed, throttle and navigation angles
QP Extended Tuning	QuadPlane VTOL PIDs and navigation (Q_ parameters), shown for Plane with QuadPlane enabled
Standard Params / Advanced Params	Parameters with descriptions, units, ranges and drop-down choices
Onboard OSD	Arrange items on the on-screen display
MAVftp	Browse files on the flight controller's SD card: download, upload, delete (logs, Lua scripts, terrain)
User Params	Your own short list of parameters to edit
Full Parameter List	Every parameter in a searchable table (chapter 11)
Airframe Limits	What the airframe can physically do, and the in-flight alarm settings (11.4, 11.5)
Planner	Program settings (below)

10.1 Planner settings

Display	Theme (Sarus Dark, Sarus Light, Sarus Glass, Mission Planner themes, Custom; a new theme applies fully after a restart) · Layout (Basic / Advanced) · UI Language · HUD options · Speed, distance and altitude units
Speech	Enable Speech; spoken mode, waypoint, battery warning, altitude, arm/disarm, 30-second interval message; Only when Armed
Map	Map Follow, map rotated to follow the plane, track length, aircraft icon, COG, heading, nav bearing, turn radius, target, ADS-B, airports, TFRs, no-fly zones, map access mode, open map cache
Telemetry Rates	Requested rates for attitude, position, mode/status, RC and sensors (lower on slow radios)
Connection	Reset on USB connect (DTR), GCS ID, params download in background, load waypoints on connect, slow-computer mode
Other	Log path · Video device · Joystick setup · Beta Updates · OptOut Anon Stats (ticked by default in Sarus) · Password Protect Config · Testing Screen · Mavlink Message Debug

11 Parameters, the lock and airframe limits

Open **CONFIG → Full Parameter List**. Every parameter is shown with its value, units, named options where it has them, and a description. Sarus takes the value you type and nothing else: it never pulls a value back into a range, and it offers no ready-made parameter sets.

11.1 Buttons

Write Params	Send all changed values (highlighted) to the aircraft
Refresh Params	Download all values from the aircraft again (F5 anywhere)
Save to file / Load from file	Save all values to a .param file; load a file (loaded values are marked as changed until you Write)
Compare Params	Compare the aircraft with a file and choose which values to take
Reset to Default	Reset all parameters to the firmware's defaults. On Sarus firmware that carries the owner's parameter set, this returns to that set (11.6). The aircraft must be recalibrated.
Commit Params	Hidden by default. Tells the aircraft to store written values permanently (same as Ctrl + Y)
Search · Modified	Filter by name or description · show only values you have changed and not yet written
Lock parameters / Unlock parameters	At the bottom of the button column. Shows whether changes are unlocked and switches between the two (11.3)

11.2 How Sarus treats your values

SITUATION	WHAT SARUS DOES
A value ArduPilot would call unusual	Accepted as typed. Sarus 1.1.0 no longer shows or applies ArduPilot's recommended ranges, and the spin box under the Options column no longer clamps to them. The only judgement on a value is your own airframe limits (11.4).
Read-only parameter	A warning asks you to confirm. If you choose Yes, the value is sent.
After Write Params	Sarus re-reads every written value from the aircraft. If the aircraft limited, rounded or rejected a value, the window Values changed by aircraft lists it, and the table shows the value the aircraft actually uses.
Suggested values	Not offered. Load Presaved and SETUP → Initial Tune Parameter , which write values the program proposes, are gone from Sarus.

WHAT THE FIRMWARE ITSELF CHANGES

We wrote every parameter of QuadPlane, Copter and Rover to the simulator and read it back, about 3,000 in all. The firmware kept a different value in only a handful of cases, and none of them is an ArduPilot preference that could be removed safely. Q_LOIT_ACC_MAX_M and LOIT_ACC_MAX_M are capped at what the maximum lean angle allows, because a multirotor tilted to 30 degrees can only accelerate sideways at $g \cdot \tan(30^\circ)$, about 5.66 m/s^2 , without losing height. EK3_BCN_DELAY, EK3_FLOW_DELAY and LAND_THR_SLEW are stored in 8 bits and stop at 127. The firmware overwrites BARO*_GND_PRESS on its own, and MIS_TOTAL cannot be written at all. Some settings, Q_ENABLE, the frame class and the serial protocols among them, only take effect after a reboot and a refresh with **F5**. In each case Sarus shows you the value the aircraft actually holds.

11.3 The parameter lock

Anyone with Sarus can connect, read every parameter, plan and fly. Changing how the aircraft is set up needs the Sarus admin password, which only the owner holds. When you try a change, Sarus asks for the password once; after that, changes are unlocked until you press **Lock parameters** or close the program.

NEEDS THE PASSWORD	STAYS OPEN
Parameter writes from any page, loading a parameter file, Reset to Default, every calibration, reboot into the bootloader, firmware install, file writes over MAVFtp, receiver pairing, Lua scripting commands, MAVLink signing setup	Connecting, reading parameters, planning and uploading missions, arming, flying, mode changes, plain reboot, downloading logs

The lock has two halves. Sarus itself refuses the change until the password is given. On aircraft running Sarus firmware (Sarus-2 or later), the aircraft refuses it as well: after the password, Sarus unlocks each connected aircraft with a signed, one-time code before the change is sent. Another ground station, an older copy of Sarus or a home-made tool cannot change the aircraft's setup without the password, even if it skips the program's own check. The aircraft reports each refusal in **Messages** as "Sarus: locked, ... refused".

Unlocking while armed	Refused by the aircraft. Unlock on the ground; an unlock made before take-off stays valid in the air, so you can tune in flight.
Link lost	An aircraft locks itself again after 10 seconds without hearing the station that unlocked it, once disarmed. It never locks in flight. A reboot always locks it.
Two stations	An unlock belongs to the station and radio link that made it. A second station, or the same station on a second radio, is still refused.
"Aircraft refused the unlock"	Its firmware was built with a different admin key. Changes to it will not take effect.
Aircraft on ArduPilot firmware	Only the program's half of the lock applies. Install Sarus firmware to lock the aircraft itself.
Lost password	Nobody can recover it. The owner keeps a spare password in a safe place for that reason. Without either, aircraft need firmware with a new key.

WHAT THE LOCK CANNOT STOP

Someone with the flight controller on a bench can still flash other firmware through its USB bootloader. Protecting against that needs a secure bootloader, which is planned. Settings inside DroneCAN peripherals are not covered either.

11.4 Airframe limits

CONFIG → Airframe Limits holds what this airframe can physically do. The page shows only the boxes that apply to the connected vehicle; leave a box empty to skip that check, and press **Save limits**. The limits are stored for this flight controller only, by its unique ID, under Documents\Sarus Operation Planner\airframes.

VEHICLE	LIMITS YOU CAN ENTER
Fixed wing and QuadPlane	Stall speed, maximum airspeed, maximum climb and descent rate, maximum bank angle, maximum nose-up and nose-down pitch
Multicopter and VTOL hover	Maximum lean angle, maximum speed, maximum climb and descent rate
Rover	Maximum speed, smallest turn radius

Any parameter that asks for more than the airframe can do blinks red: in the Full Parameter List as soon as you type it, and in a banner across the top of the window for as long as the aircraft holds it. The banner names the parameter and the limit; **Details** lists every one, and **Hide** hides it until the values change again. Sarus never changes or refuses the value. It understands the parameter names of both ArduPilot 4.6 and 4.7, in their own units (centidegrees, cm/s, degrees or m/s).

11.5 The in-flight alarm

During a mission (AUTO or GUIDED, armed), Sarus compares what the aircraft is trying to do with what it achieves: altitude, airspeed and position on the planned track. When one of them stays further off than the alarm settings on the Airframe Limits page, for longer than the set time, and is not getting smaller, a siren sounds and a window offers three choices.

Continue plan	The mission goes on unchanged. The alarm stays quiet for 60 seconds.
Hold position	The aircraft loiters where it is (QLOITER when hovering, HOLD for a rover).
Return home	The aircraft returns to launch (RTL).
No answer in 15 seconds	The mission continues with the parameters as set. The alarm never changes anything on its own.

Defaults: altitude 15 m, airspeed 4 m/s, track 40 m, for 10 seconds. Take-off, landing and return legs are not judged, nor is airspeed before the aircraft is flying on its wings, because large differences there are normal. An error that is closing, such as a climb to a new altitude, does not raise the alarm. In the simulator, the alarm stayed silent through take-off, transition and climb, and sounded 11 seconds after the engine was cut in cruise.

11.6 Sarus firmware and its defaults

Sarus firmware comes in two lines with the same Sarus changes: one on ArduPilot 4.7.1, the newest stable release, and one on 4.6.3, the last release of the longer-proven 4.6 series. Install Firmware lists both and asks which to load. CubeOrangePlus is offered on the 4.7.1 line only, because ArduPilot 4.6.3 leaves too little flash on that board for the lock. Firmware built with the owner's parameter set uses it as its defaults, so a new board, a parameter reset or a firmware install starts from the owner's values. Sensor calibrations are never part of that set; every aircraft still needs its own.

GOOD PRACTICE

Save to file before you change anything. Change a few values at a time. After a tuning session, save again with a descriptive file name and date, and press Lock parameters before you hand the laptop to someone else.

12 SIMULATION

The simulator runs ArduPilot's real firmware (SITL) on your PC, so you can fly without any hardware. We use it for training, for rehearsing missions and for trying out parameter changes before they go near an aircraft.

- 1 Drag the **Home Location** marker to where you want to fly, and set the heading.
- 2 Choose a vehicle: **Plane**, **Multirotor**, **Helicopter** or **Rover**. For a QuadPlane, open the advanced options and choose a quadplane model.
- 3 The firmware is downloaded and started; the program connects automatically.

Advanced options: **Model**, **Sim Speed**, **Wipe** (reset parameters), **Extra command line**, and swarm modes (several simulated aircraft).

13 HELP and updates

Check for Updates · **Check for BETA Updates** · **Change Log** (opens the Sarus release notes) · **Show Console Window (restart)** (shows diagnostic output, useful for support).

14 Keyboard shortcuts

Every shortcut works exactly as in Mission Planner. The list below was taken from the Sarus source code, and the automated tests press each one.

Anywhere in the program

KEYS	ACTION
F2	DATA (Flight Data) screen
F3	PLAN (Flight Planner) screen
F4	CONFIG (Software Config) screen
F5	Refresh all parameters from the aircraft and reload the current screen
F12	Connect / disconnect
Ctrl + F	Advanced tools window (special functions and test tools)
Ctrl + P	Plugins: enable or disable plugins
Ctrl + G	NMEA output: send the aircraft position as GPS NMEA to a serial port
Ctrl + X	Map cache manager
Ctrl + L	Spectrogram (vibration / FFT view)
Ctrl + W	Radio propagation (coverage) settings
Ctrl + Z	Camera test (outside PLAN; on PLAN it is Undo)
Ctrl + T	Connect without downloading parameters
Ctrl + Y	Tell the aircraft to save its parameters to permanent storage
Ctrl + J	DevOps window (developer diagnostics)

DATA screen

Ctrl + 1 ... 0	Select tab 1 to 10
Space	Play / pause log playback
Keypad + / -	Faster / slower playback
Ctrl +click map	Fly To Here

PLAN screen

Ctrl + Z	Undo
Ctrl + O / S	Open / save mission file
Ctrl + Shift + W	Write mission
Ctrl + Shift + F	Write Fast
Ctrl + Shift + R	Read mission
Ctrl + N	Toggle map drawing mode (GDI)
Ctrl + M	Copy the plan into the program's mission memory (no upload)
Ctrl +drag / click	Select several waypoints

15 QuadPlane (VTOL) quick-start workflow

A suggested order for setting up a new QuadPlane. Props off until step 10.

- 1 **Firmware:** SETUP → Install Firmware → Plane.
- 2 **Connect** over USB (115200) and let parameters download.
- 3 **Enable QuadPlane:** set Q_ENABLE = 1 in the Full Parameter List, Write, reboot the board, press **F5**. Set Q_FRAME_CLASS and Q_FRAME_TYPE, Write, reboot.
- 4 **Outputs:** SETUP → Servo Output: assign motors 1–4, aileron, elevator, rudder and throttle functions; check reverse settings.
- 5 **Calibrate:** Accel Calibration, Compass, Radio Calibration. Calibrate the airspeed sensor (Optional Hardware → Airspeed).
- 6 **Flight modes:** include QHOVER, QLOITER, FBWA and a return mode (RTL/QRTL).
- 7 **Failsafes:** radio, battery and GCS failsafes, with VTOL-appropriate actions.
- 8 **Motor Test** at 5%: order and direction for each VTOL motor; check surfaces move correctly in FBWA by hand.
- 9 **Save a parameter file** as your baseline.
- 10 **First flight:** follow the Sarus Field Test Checklist (QHOVER → QLOITER → transition → TECS check → small AUTO mission with VTOL_TAKEOFF and VTOL_LAND).
- 11 **Tune** with CONFIG → Extended Tuning (TECS), QP Extended Tuning, and the Full Parameter List; review logs after each flight.

Key QuadPlane and TECS parameters

PARAMETER	MEANING
Q_ENABLE	Turn QuadPlane support on
Q_FRAME_CLASS / Q_FRAME_TYPE	VTOL motor layout
Q_ASSIST_SPEED	Below this airspeed, VTOL motors help in fixed-wing modes
ARSPD_FBW_MIN / ARSPD_FBW_MAX	Minimum and maximum airspeed in fixed-wing flight
Q_TRANSITION_MS	Time to transition from VTOL to forward flight
TECS_CLMB_MAX	Maximum climb rate (m/s)
TECS_SINK_MIN / TECS_SINK_MAX	Minimum and maximum sink rate (m/s)
TECS_TIME_CONST	How quickly TECS corrects height and speed errors
TECS_SPDWEIGHT	Priority between speed and height
Q_RTL_MODE, Q_OPTIONS	VTOL return and landing behaviour, options

Parameter names change between ArduPilot versions. Always check the description shown in the Full Parameter List for your firmware.

16 Logs and files

WHAT	WHERE
Telemetry logs (.tlog, .rlog)	Documents\Sarus Operation Planner\logs\<vehicle>\<id>\, one per connection, recorded automatically
Downloaded DataFlash logs (.bin)	Same logs folder
Program settings	Documents\Sarus Operation Planner\config.xml
Map cache, terrain (SRTM)	Documents\Sarus Operation Planner\gmapcache, \srtm
Program diagnostic log	C:\ProgramData\Sarus Operation Planner\SarusOperationPlanner.log
Program files	C:\Program Files (x86)\Sarus Operation Planner\

The log folder can be changed in CONFIG → Planner → Log Path. The installer registers .tlog, .bin and .log files, so you can open them with Sarus Operation Planner from Windows Explorer (right-click → Open with).

Telemetry logs and DataFlash logs

The **telemetry log** (.tlog) holds what the ground station received over the radio and can be replayed on the DATA screen. The flight controller writes its own **DataFlash log** (.bin), at full rate and with far more detail; download it after flight (DATA → DataFlash Logs) and open it with **Review a Log** to plot values, or use **Auto Analysis**.

17 Troubleshooting

PROBLEM	WHAT TO CHECK
No COM port in the list	Cable is a data cable; drivers installed (reinstall Sarus); try another USB port; check Windows Device Manager.
"No heartbeat" / connection timeout	Correct port and speed (USB 115200, SiK 57600); the port is not used by Mission Planner or another program; radios are paired and powered.
Parameter download stops	Weak radio link: move closer or use USB, or reduce telemetry rates (CONFIG → Planner). Press F5 to retry.
Cannot arm	Read the PreArm message on the HUD or in the Messages tab and fix its cause; never disable arming checks to fly.
"Values changed by aircraft"	The firmware limited or rejected a value, or the aircraft is locked. The table shows the real value. See 11.2 and 11.3.

PROBLEM	WHAT TO CHECK
Password asked for every change	Expected while locked. Type the admin password once; changes stay unlocked until you press Lock parameters or close Sarus.
"Sarus: locked, ... refused" in Messages	The aircraft refused a change because this station has not unlocked it. Unlock again while disarmed; check that only one station is connected.
Red banner at the top	A parameter asks for more than the airframe limits allow. Press Details, then fix the parameter or the limit (11.4).
Firmware page empty	Internet connection needed; disconnect from the aircraft first.
Map is grey	No internet and no cached tiles for that area. Prefetch maps before going to the field (PLAN → right-click → Map Tool → Prefetch).
"Windows protected your PC"	See 4.2. Appears until the installer is code-signed.
Blurry text	Windows display scaling above 100%. Same as Mission Planner.
Program closes or freezes	Send the diagnostic log (chapter 16) and the telemetry log to Sarus support.

18 Privacy, licence and support

Network connections the program makes

ArduPilot servers	Firmware lists and downloads, parameter descriptions, SITL binaries
GitHub (Sarus releases)	Update checks and update downloads
Map providers	Map tiles and elevation data for the area you view (same providers as Mission Planner)
Optional plugins	Only if you enable them (for example internet ADS-B traffic)
Usage statistics	Off by default. Nothing is sent unless you turn it on.

Telemetry, logs, missions and parameters stay on your computer unless you share them.

Licence

Sarus Operation Planner is free software under the GNU General Public License version 3. You may copy, change and redistribute it under the terms of that licence. It is provided **without any warranty**. Complete source code: github.com/LikhitS/SarusOperationPlanner.

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Support

Sarus Aerospace Pvt Ltd · www.sarusaerospace.in. When reporting a problem, include the Sarus version (HELP screen), firmware version, the steps to reproduce it, the diagnostic log and the telemetry log.

19 Abbreviations

ADS-B	Automatic Dependent Surveillance–Broadcast (aircraft position broadcasts)
AHRS	Attitude and Heading Reference System
CAN	Controller Area Network (DroneCAN / UAVCAN peripherals)
COG	Course Over Ground
DGCA	Directorate General of Civil Aviation (India)
EKF	Extended Kalman Filter (position and attitude estimator)
ESC	Electronic Speed Controller
FBWA / FBWB	Fly-By-Wire A / B (assisted fixed-wing modes)
FFT	Fast Fourier Transform (vibration analysis, harmonic notch)
GCS	Ground Control Station
GPL	GNU General Public License
HDOP	Horizontal Dilution of Precision (GPS accuracy, lower is better)
HUD	Head-Up Display
KML / SHP	Google Earth file / shapefile (map shapes)
MAVLink	Micro Air Vehicle Link (aircraft–ground protocol)
MSI	Microsoft Installer package
NMEA	GPS sentence format (National Marine Electronics Association)
OSD	On-Screen Display (video overlay)
PID	Proportional–Integral–Derivative control loop
POI / ROI	Point / Region Of Interest
PWM	Pulse Width Modulation (servo and RC signals, in microseconds)
QHOVER / QLOITER / QLAND / QRTL	QuadPlane VTOL modes: hover, hold position, land, return to launch
RC	Radio Control
REPL	Read–Eval–Print Loop (interactive script console)
RTCM / RTK	GPS correction format / Real-Time Kinematic GPS
RTL	Return To Launch
SiK	Common open-source telemetry radio firmware
SITL	Software In The Loop (simulator)
SRTM	Shuttle Radar Topography Mission (terrain elevation data)
TECS	Total Energy Control System (ArduPlane speed and height control)
tlog / bin	Telemetry log / DataFlash (on-board) log
UTM	Universal Transverse Mercator (grid coordinates)
VTOL	Vertical Take-Off and Landing
WP	Waypoint